Youssef Hesham Mahmoud SE3

Lab 01

Part A: VM vs Container Resource Comparison

Virtual Machine (Ubuntu 22.04 — Hyper-V)

Memory & process usage inside the VM:

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings


To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ubuntu-vm:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           891Mi       201Mi       135Mi       0.0Ki       555Mi       546Mi
Swap:             0B          0B          0B
ubuntu@ubuntu-vm:~$ ps aux --sort=-%mem | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         733  0.0  4.4 1858208 40820 ?        Ssl  Apr23   0:02 /snap/snapd/current/usr/lib/snapd/snapd
root         410  0.0  2.9 289352 27100 ?        SLsl Apr23   0:00 /sbin/multipathd -d -s
root         791  0.0  2.3 110156 21600 ?        Ssl  Apr23   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
gnal
root        2210  0.0  2.2 296024 20280 ?        Ssl  Apr23   0:00 /usr/libexec/packagekitd
root         727  0.0  2.0  33076 19084 ?        Ss   Apr23   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
systemd+     603  0.0  1.5  26468 14092 ?        Ss   Apr23   0:00 /lib/systemd/systemd-resolved
root           1  0.0  1.4 102032 12992 ?        Ss   Apr23   0:00 /sbin/init
root        2069  0.0  1.1  16924 10684 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
root        1353  0.0  1.1  16928 10568 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
ubuntu@ubuntu-vm:~$ ps aux --sort=-%mem | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         733  0.0  4.4 1858208 40820 ?        Ssl  Apr23   0:02 /snap/snapd/current/usr/lib/snapd/snapd
root         410  0.0  2.9 289352 27100 ?        SLsl Apr23   0:00 /sbin/multipathd -d -s
root         791  0.0  2.3 110156 21600 ?        Ssl  Apr23   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
gnal
root        2210  0.0  2.2 296024 20280 ?        Ssl  Apr23   0:00 /usr/libexec/packagekitd
root         727  0.0  2.0  33076 19084 ?        Ss   Apr23   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
systemd+     603  0.0  1.5  26468 14092 ?        Ss   Apr23   0:00 /lib/systemd/systemd-resolved
root           1  0.0  1.4 102032 12992 ?        Ss   Apr23   0:00 /sbin/init
root        2069  0.0  1.1  16924 10684 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
root        1353  0.0  1.1  16928 10568 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
ubuntu@ubuntu-vm:~$
```

```
ubuntu@ubuntu-vm:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            90M  916K   89M   2% /run
/dev/sda1       4.7G  1.7G  3.1G  35% /
tmpfs           446M     0  446M   0% /dev/shm
tmpfs           5.0M     0  5.0M   0% /run/lock
/dev/sda15      105M  6.1M   99M   6% /boot/efi
tmpfs            90M  4.0K   90M   1% /run/user/1000
ubuntu@ubuntu-vm:~$ exit
logout

C:\Users\PC>docker run -it --name test-container ubuntu /bin/bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
b40150c1c271: Pull complete
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b
Status: Downloaded newer image for ubuntu:latest
root@f9dbb2bdd720:/# free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       922Mi       6.4Gi       9.6Mi       549Mi       6.7Gi
Swap:          2.0Gi          0B       2.0Gi
root@f9dbb2bdd720:/# ps aux --sort=-%mem | head
USER       PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root        10  0.0  0.0   7888  4044 pts/0    R+   23:04   0:00 ps aux --sort=-%mem
root         1  0.0  0.0   4588  4000 pts/0    Ss   23:04   0:00 /bin/bash
root        11  0.0  0.0   2708  1092 pts/0    S+   23:04   0:00 head
root@f9dbb2bdd720:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay        1007G  2.8G  953G   1% /
tmpfs            64M     0   64M   0% /dev
tmpfs           3.9G     0  3.9G   0% /sys/fs/cgroup
shm              64M     0   64M   0% /dev/shm
/dev/sdd       1007G  2.8G  953G   1% /etc/hosts
tmpfs           3.9G     0  3.9G   0% /proc/acpi
tmpfs           3.9G     0  3.9G   0% /sys/firmware
root@f9dbb2bdd720:/#
```

Docker Container

```
C:\Users\PC>docker run -it --name test-container ubuntu /bin/bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
b40150c1c271: Pull complete
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41
c7b
Status: Downloaded newer image for ubuntu:latest
root@f9dbb2bdd720:/# free -h
               total        used        free      shared  buff/cache   avail
able
Mem:           7.6Gi       922Mi       6.4Gi       9.6Mi       549Mi       6
.7Gi
Swap:          2.0Gi          0B       2.0Gi
root@f9dbb2bdd720:/# ps aux --sort=-%mem | head
USER        PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         10  0.0  0.0   7888  4044 pts/0    R+   23:04   0:00 ps aux --so
rt=-%mem
root          1  0.0  0.0   4588  4000 pts/0    Ss   23:04   0:00 /bin/bash
root         11  0.0  0.0   2708  1092 pts/0    S+   23:04   0:00 head
root@f9dbb2bdd720:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay        1007G  2.8G  953G   1% /
tmpfs            64M     0   64M   0% /dev
tmpfs           3.9G     0  3.9G   0% /sys/fs/cgroup
shm              64M     0   64M   0% /dev/shm
/dev/sdd       1007G  2.8G  953G   1% /etc/hosts
tmpfs           3.9G     0  3.9G   0% /proc/acpi
tmpfs           3.9G     0  3.9G   0% /sys/firmware
root@f9dbb2bdd720:/# exit
exit
```

Comparison Table

Metric	Virtual Machine	Docker Container
Total RAM	891 MB	7.6 GB (shared host)
Used RAM	201 MB	922 MB
Active Processes	10+ system processes	3 processes only
Disk Size	4.7 GB	1007 GB (shared host)
Startup Time	Several seconds	Near instant
Isolation	Full OS isolation	Process-level isolation

Part B: Cloud Infrastructure Exploration

Hypervisor Detection (dmesg)

```
C:\Users\PC>multipass shell ubuntu-vm
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-173-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Apr 23 01:42:03 EET 2026

  System load:  0.0               Processes:             97
  Usage of /:   34.3% of 4.67GB   Users logged in:       0
  Memory usage: 28%               IPv4 address for eth0: 172.27.100.105
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings


Last login: Thu Apr 23 01:42:40 2026 from 172.27.96.1
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ubuntu-vm:~$ dmesg | grep -i hypervisor
dmesg: read kernel buffer failed: Operation not permitted
ubuntu@ubuntu-vm:~$ sudo dmesg | grep -i hypervisor
[    0.000000] Hypervisor detected: Microsoft Hyper-V
ubuntu@ubuntu-vm:~$ sudo dmesg | grep -i hypervisor
[    0.000000] Hypervisor detected: Microsoft Hyper-
[    0.000000] Hypervisor detected: Microsoft Hyper-V
```

Result: Hypervisor detected: Microsoft Hyper-V — same role as AWS Nitro Hypervisor.

System Information (dmidecode)

```
ubuntu@ubuntu-vm:~$ sudo dmidecode | grep -A3 "System Information"
System Information
        Manufacturer: Microsoft Corporation
        Product Name: Virtual Machine
        Version: Hyper-V UEFI Release v4.1
ubuntu@ubuntu-vm:~$
```

Manufacturer: Microsoft Corporation | Product: Virtual Machine | Version: Hyper-V UEFI Release v4.1

Process Monitoring — top

```
ubuntu@ubuntu-vm:~$ top
top - 06:34:56 up  2:24,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  89 total,   1 running,  88 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    892.0 total,    136.1 free,    191.3 used,    564.6 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    556.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  102032  12992   8384 S   0.0   1.4   0:01.27 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     11 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_tasks_rude_
     12 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_tasks_trace
     13 root      20   0       0      0      0 S   0.0   0.0   0:00.10 ksoftirqd/0
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.27 rcu_sched
     15 root      rt   0       0      0      0 S   0.0   0.0   0:00.05 migration/0
     16 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     20 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 khungtaskd
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     24 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 writeback
     25 root      20   0       0      0      0 S   0.0   0.0   0:00.27 kcompactd0
     26 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     27 root      39  19       0      0      0 S   0.0   0.0   0:00.00 khugepaged
     73 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kintegrityd
     74 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kblockd
     75 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 blkcg_punt_bio
     76 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 tpm_dev_wq
     77 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 ata_sff
     78 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 md
     79 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 edac-poller
     80 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 devfreq_wq
```

Process Monitoring — htop

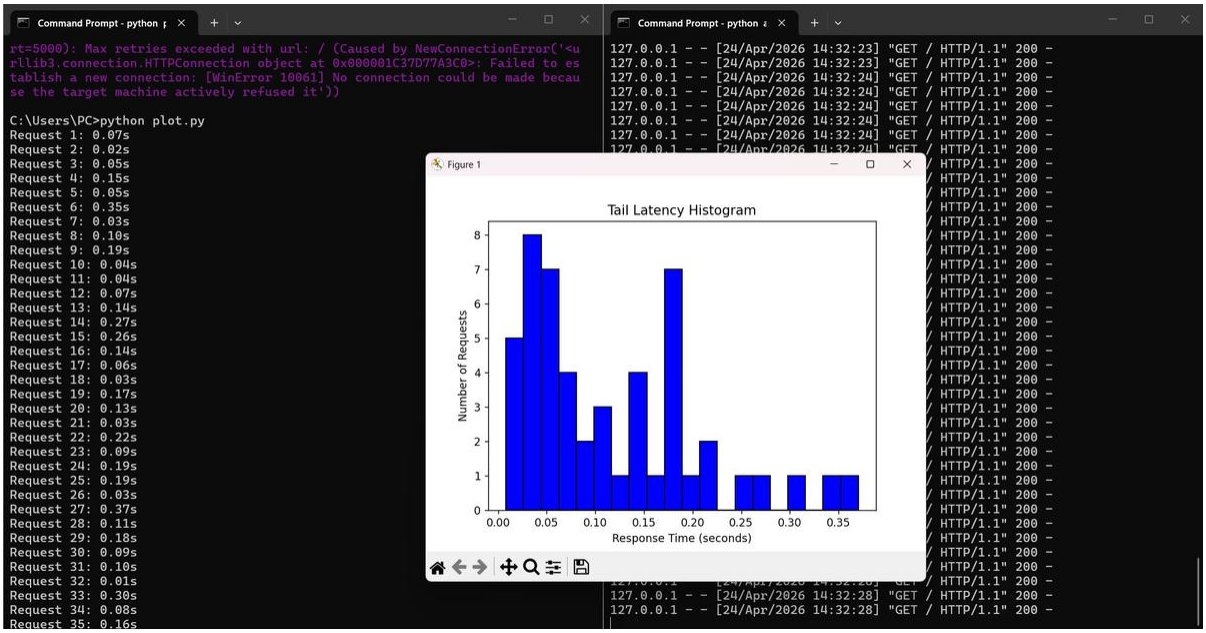
```
  CPU[|                                                        1.3%]   Tasks: 26, 24 thr; 1 running
  Mem[||||||||||||||||||||||||||||||||||||||||||||||||||||191M/892M]   Load average: 0.00 0.00 0.00
  Swp[                                                       0K/0K]   Uptime: 02:25:48

    PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
    417 root       RT   0  282M 27100  9072 S  0.0  3.0  0:00.55 /sbin/multipathd -d -s
      1 root       20   0   99M 12992  8384 S  0.0  1.4  0:01.27 /sbin/init
    372 root       19  -1 47892 10476  9340 S  0.0  1.1  0:00.14 /lib/systemd/systemd-journald
    410 root       RT   0  282M 27100  9072 S  0.0  3.0  0:00.65 /sbin/multipathd -d -s
    412 root       20   0 23056  6432  4660 S  0.0  0.7  0:00.73 /lib/systemd/systemd-udevd
    413 root       20   0  282M 27100  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
    414 root       RT   0  282M 27100  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
    415 root       RT   0  282M 27100  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
    416 root       RT   0  282M 27100  9072 S  0.0  3.0  0:00.01 /sbin/multipathd -d -s
    418 root       RT   0  282M 27100  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
    542 systemd-t  20   0 89364  6636  5832 S  0.0  0.7  0:00.06 /lib/systemd/systemd-timesyncd
    548 systemd-t  20   0 89364  6636  5832 S  0.0  0.7  0:00.00 /lib/systemd/systemd-timesyncd
    601 systemd-n  20   0 16128  7976  6976 S  0.0  0.9  0:00.06 /lib/systemd/systemd-networkd
    603 systemd-r  20   0 26468 14092  8944 S  0.0  1.5  0:00.23 /lib/systemd/systemd-resolved
    718 root       20   0  7288  2732  2496 S  0.0  0.3  0:00.01 /usr/sbin/cron -f -P
    719 messagebu  20   0  8660  4644  3956 S  0.0  0.5  0:00.06 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog
    727 root       20   0 33076 19084 10236 S  0.0  2.1  0:00.05 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
    729 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.02 /usr/sbin/rsyslogd -n -iNONE
    733 root       20   0 1814M 41204 25772 S  0.0  4.5  0:03.47 /snap/snapd/current/usr/lib/snapd/snapd
    734 root       20   0 15532  7376  6420 S  0.0  0.8  0:00.05 /lib/systemd/systemd-logind
    758 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.01 /usr/sbin/rsyslogd -n -iNONE
    759 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.00 /usr/sbin/rsyslogd -n -iNONE
    760 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.00 /usr/sbin/rsyslogd -n -iNONE
    784 root       20   0  6176  1092  1004 S  0.0  0.1  0:00.00 /sbin/agetty -o -p -- \u --noclear tty1 linux
    791 root       20   0  107M 21600 13484 S  0.0  2.4  0:00.07 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-sign
    813 root       20   0  107M 21600 13484 S  0.0  2.4  0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-sign
    818 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.33 /snap/snapd/current/usr/lib/snapd/snapd
    819 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
    820 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
    827 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.19 /snap/snapd/current/usr/lib/snapd/snapd
    828 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.75 /snap/snapd/current/usr/lib/snapd/snapd
    879 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.30 /snap/snapd/current/usr/lib/snapd/snapd
    881 root       20   0 1814M 41204 25772 S  0.7  4.5  0:00.75 /snap/snapd/current/usr/lib/snapd/snapd
    883 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

The VM shows 89 total tasks, 1 running, 88 sleeping. Memory: 892 MB total, 191 MB used.

Part C: Tail Latency Mini-Simulation

Flask App Response Times & Histogram



Most requests complete in 0.02–0.07s, but a small number take 0.25–0.35s — this is the tail latency.

- **Short Written Answers / Reflection**

1. **Difference between VM and Container**
   - **VM runs a full operating system and uses more resources**
   - **Container shares host OS and is lightweight**
   - **Containers start much faster than VMs**
2. **Resource Overhead**
   - **VM consumes more memory and CPU**
   - **Container is efficient and uses fewer resources**

3. **AWS Nitro Hypervisor**

   - **Nitro is a secure hypervisor used by AWS**
   - **It is visible in the system but cannot be accessed**
   - **Provides strong isolation and security**

4. **Tail Latency**

   - **Tail latency refers to slow response times at the end of distribution**
   - **Even if most requests are fast, some are very slow**
   - **This affects user experience in real systems**